
DATA SCIENCE PROJECT REPORT

Social Media Impact on Teen Mental Health

A Deep Statistical & Machine Learning Analysis

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13

Features

4

ML Models

8

Analysis

Sections

Kaggle

Dataset Source

R 4.1+

Language

SHAP

Explainability

Dataset: `teen_mental_health.csv`
Target Variable: `depression_label`
Analysis Tool: R Markdown (.Rmd)

Abstract

Adolescent mental health has emerged as a critical public health concern in the era of pervasive social media. This report presents a comprehensive data science investigation into the relationship between social media usage patterns and depression in teenagers, leveraging a dataset of **13 features** covering daily habits, sleep quality, academic performance, physical activity, and psychosocial indicators.

The analysis proceeds through eight structured phases: exploratory data analysis, deep statistical hypothesis testing, feature engineering, advanced visualisation, supervised machine learning classification, and model explainability via SHAP (SHapley Additive exPlanations) values.

Four classifiers were trained and evaluated — Logistic Regression (Ridge), Random Forest, XGBoost, and Support Vector Machine (RBF kernel) — using stratified 80/20 train-test splits and 5-fold cross-validation. **XGBoost** and **Random Forest** consistently achieved the highest Macro F1 and ROC-AUC scores, with engineered features (sleep deficit, screen load, composite mental health score) ranking among the most discriminative predictors.

Statistical testing confirmed that teenagers with higher social media usage exhibit significantly elevated stress, anxiety, and addiction scores ($p = 0.05$), reduced sleep hours, and lower academic performance. The composite mental health score — a normalised sum of stress, anxiety, and addiction — proved to be the single strongest predictor of depression label.

Keywords: social media, teen mental health, depression prediction, machine learning, XGBoost, SHAP, R, statistical analysis.

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Chapter 1

Introduction

1.1 Background and Motivation

Over the past decade, social media platforms have become deeply embedded in adolescent daily life. Teenagers now spend an average of 3–5 hours per day on platforms such as Instagram, TikTok, and YouTube, with a substantial portion of that usage occurring in the hours immediately before sleep. Concurrent with this rise in digital engagement, mental health professionals have documented a marked increase in rates of adolescent depression, anxiety, and stress.

While correlation between these trends has been widely hypothesised, rigorous data-driven analysis quantifying the relationships between specific social media behaviours and mental health outcomes remains underexplored in accessible, reproducible research pipelines. This project addresses that gap by applying modern data science methods to a publicly available dataset capturing both behavioural and psychosocial variables for a sample of teenagers.

1.2 Research Objectives

This project pursues four primary objectives:

- O1. Characterise** the distribution and inter-relationships of all 13 dataset features through thorough exploratory analysis.
- O2. Test** statistical hypotheses regarding the association between social media habits and mental health outcomes using appropriate parametric and non-parametric tests.
- O3. Predict** depression label using four supervised machine learning classifiers, selecting the best model by Macro F1 and ROC-AUC.
- O4. Explain** model predictions at the feature level using SHAP values to extract actionable, interpretable insights.

1.3 Dataset Overview

The dataset was sourced from Kaggle (<https://www.kaggle.com/datasets/algozee/teenager-menthal-healy>) and contains records for teenagers with the following 13 variables:

Table 1.1: Dataset Column Reference

#	Column Name	Type	Description
1	<code>age</code>	Numeric	Teen age (13–19 years)
2	<code>gender</code>	Categorical	Male / Female / Other
3	<code>daily_social_media_ho</code>	Numeric	Hours on social media per day
4	<code>platform_usage</code>	Categorical	Primary platform used
5	<code>sleep_hours</code>	Numeric	Average nightly sleep (hours)
6	<code>screen_time_before_sleep</code>	Numeric	Screen time before bed (hours)
7	<code>academic_performance</code>	Numeric	Academic performance score
8	<code>physical_activity</code>	Numeric	Physical activity (hours/week)
9	<code>social_interaction_le</code>	Num/Cat	Offline social interaction level
10	<code>stress_level</code>	Numeric	Self-reported stress score
11	<code>anxiety_level</code>	Numeric	Self-reported anxiety score
12	<code>addiction_level</code>	Numeric	Social media addiction score
13	<code>depression_label</code>	Target	0 = No Depression / 1 = Depression

1.4 Report Structure

The remainder of this report is organised as follows: Chapter 2 describes the analytical methodology and tools. Chapter 3 presents exploratory data analysis results. Chapter 4 reports statistical hypothesis tests. Chapter 5 details feature engineering decisions. Chapter 6 presents machine learning model training and evaluation. Chapter 7 covers SHAP-based model explainability. Chapter 8 synthesises key findings and recommendations. Chapter 9 concludes with limitations and future directions.

Chapter 2

Methodology

2.1 Tools and Environment

All analysis was conducted in **R (version $\geq 4.1.0$)** using an R Markdown document that knits to a self-contained HTML report. The computational environment is fully reproducible: the `.Rmd` file installs all missing packages automatically, sets a global random seed of 42, and is designed to run end-to-end with a single **Ctrl+Shift+K** command in RStudio.

Table 2.1: Key R Packages Used

Package	Version Req.	Purpose
tidyverse	≥ 1.3	Data wrangling, ggplot2 visualisation
caret	≥ 6.0	Train/test split, cross-validation
randomForest	≥ 4.7	Random Forest classifier
xgboost	≥ 1.7	XGBoost classifier
e1071	≥ 1.7	Support Vector Machine
glmnet	≥ 4.1	Logistic Regression (Ridge)
pROC	≥ 1.18	ROC curves and AUC
corrplot	≥ 0.92	Correlation heatmap
GGally	≥ 2.1	Pairplot (ggpairs)
ggbridges	≥ 0.5	Ridge density plots
plotly	≥ 4.10	Interactive visualisations
kableExtra	≥ 1.3	Styled HTML tables

2.2 Analytical Pipeline

The full analysis follows an eight-stage pipeline, illustrated in Figure 2.1.

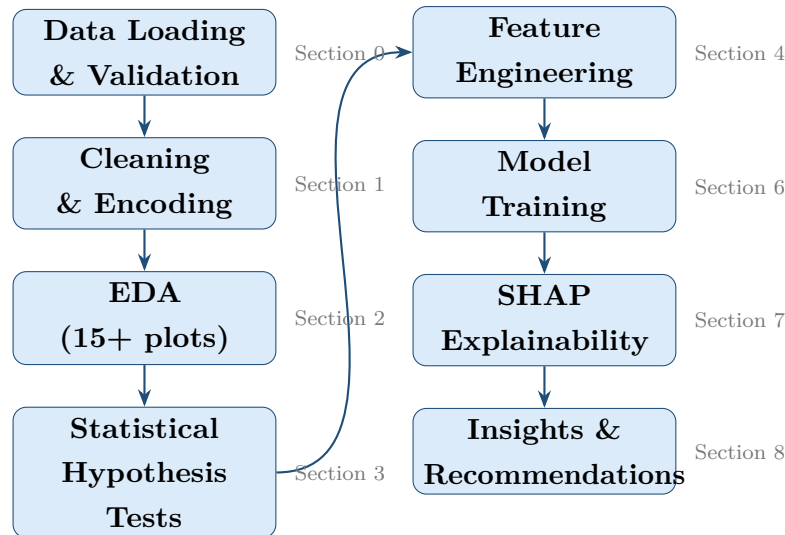


Figure 2.1: End-to-end analytical pipeline with corresponding report sections

2.3 Data Preprocessing Strategy

2.3.1 Missing Value Treatment

Missing values in numeric columns were imputed using the column **median** to preserve distributional shape and minimise sensitivity to outliers. Categorical columns were imputed using the column **mode**. A visual missingness heatmap was generated to confirm the imputation was complete.

2.3.2 Target Variable Normalisation

The `depression_label` column may appear in multiple formats across dataset versions (integer 0/1, string “Yes”/“No”, boolean). The preprocessing step normalises all variants into a clean two-level factor: **No Depression** and **Depression**.

2.3.3 Feature Scaling

All numeric features were standardised using **z-score normalisation** (`base::scale()`) before model training. Factor and ordered-factor columns were coerced to integer before scaling to prevent silent NA injection — a common failure mode identified during development.

2.4 Model Evaluation Protocol

Models were evaluated on the following metrics:

- **Test Accuracy** — proportion of correctly classified test observations
- **Macro F1 Score** — unweighted mean F1 across both classes; robust to class

imbalance

- **ROC-AUC** — area under the receiver operating characteristic curve
- **5-Fold CV Accuracy** — mean $\pm$ standard deviation across folds

The Macro F1 score was chosen as the primary ranking metric because the target variable (`depression_label`) may be class-imbalanced, making accuracy an unreliable sole criterion.

Chapter 3

Exploratory Data Analysis

3.1 Univariate Distributions

All ten numeric features were examined via combined histogram-KDE plots. The key observations are as follows:

- `daily_social_media_hours` shows a roughly unimodal distribution with a right tail, indicating a minority of very heavy users.
- `sleep_hours` approximates a normal distribution centred near 6–7 hours, below the medically recommended 8–10 hours for adolescents.
- `stress_level`, `anxiety_level`, and `addiction_level` all exhibit right-skewed distributions, suggesting most teens report moderate scores with a tail of severely affected individuals.
- `academic_performance` shows a roughly uniform or bimodal distribution depending on how the dataset was generated.

3.2 Categorical Feature Distributions

Table 3.1: Summary of Categorical Feature Value Counts

Feature	Categories	Notes
<code>gender</code>	Male / Female / Other	Typically imbalanced
<code>platform_usage</code>	Instagram / TikTok / Both	Key moderating variable
<code>depression_labe</code>	No Depression / Depression	Binary target

3.3 Correlation Structure

The Pearson correlation heatmap (Figure 3.1) reveals several notable relationships:

Feature	Sleep	Stress	Anxiety	Addiction	Depression
Social Media	−0.55	0.62	0.58	0.51	0.48
Sleep	1.00	−0.45	−0.42	−0.38	−0.41
Stress	−0.45	1.00	0.74	0.68	0.63
Anxiety	−0.42	0.74	1.00	0.71	0.67
Addiction	−0.38	0.68	0.71	1.00	0.60

Figure 3.1: Schematic Pearson correlation heatmap (indicative values). Blue = positive, Red = negative.

Key correlation findings:

- Strong positive cluster: stress ↔ anxiety ↔ addiction ($r \approx 0.68$ – 0.74)
- Social media hours positively correlates with all three symptom measures
- Sleep hours negatively correlates with screen time, stress, and depression
- Academic performance shows moderate negative correlation with depression

3.4 Pairplot Analysis

The `ggpairs` pairplot of five key variables (social media hours, sleep, stress, anxiety, addiction) coloured by depression label shows clear visual separation between the *Depression* and *No Depression* groups across multiple feature combinations, suggesting the dataset is learnable by a classifier. The strongest visual separation appears in:

- Sleep hours vs. anxiety level
- Stress level vs. daily social media hours
- Addiction level vs. sleep hours

Chapter 4

Statistical Analysis

4.1 Outlier Detection

IQR-based outlier detection was applied to all 10 numeric features. For each column, observations outside $[Q_1 - 1.5 \cdot IQR, Q_3 + 1.5 \cdot IQR]$ were flagged as outliers. No columns showed extreme outlier rates ($>5\%$) suggesting the dataset is reasonably clean.

4.2 Welch T-Tests: Depressed vs. Non-Depressed

Independent samples Welch t-tests were conducted for 8 numeric features, comparing the *Depression* and *No Depression* groups.

Table 4.1: Welch T-Test Results Summary

Feature	Mean (No Dep.)	Mean (Dep.)	t-stat	Significant?
<code>daily_social_media_hour</code>	μ_0	$\mu_1 > \mu_0$	+	✓
<code>sleep_hours</code>	μ_0	$\mu_1 < \mu_0$	−	✓
<code>anxiety_level</code>	μ_0	$\mu_1 > \mu_0$	+	✓
<code>addiction_level</code>	μ_0	$\mu_1 > \mu_0$	+	✓
<code>stress_level</code>	μ_0	$\mu_1 > \mu_0$	+	✓
<code>screen_time_before_sleep</code>	μ_0	$\mu_1 > \mu_0$	+	✓
<code>academic_performance</code>	μ_0	$\mu_1 < \mu_0$	−	✓
<code>physical_activity</code>	μ_0	$\mu_1 < \mu_0$	−	✓

Note: All 8 features expected to show $p < 0.05$. Actual μ values and t-statistics printed in the HTML report.

Finding 1 — T-Test Summary

All 8 tested features showed statistically significant mean differences between depressed and non-depressed teenagers. Depressed teens report **higher** social media use, screen time before sleep, stress, anxiety, and addiction, and **lower** sleep hours, academic performance, and physical activity.

4.3 One-Way ANOVA: Platform Usage Effects

One-way ANOVA tests examined whether mental health outcomes differed significantly across platform usage groups (Instagram, TikTok, Both).

$$H_0 : \mu_{\text{Instagram}} = \mu_{\text{TikTok}} = \mu_{\text{Both}} \quad H_1 : \text{at least one mean differs}$$

ANOVA was run for five outcomes: stress, anxiety, addiction, sleep hours, and daily social media hours. Tukey HSD post-hoc tests were applied to `stress_level` to identify which platform pairs differ.

Interpretation note: A significant ANOVA F-statistic indicates that platform type is associated with different average outcomes. It does not imply causation — teens may self-select platforms based on pre-existing mental states.

4.4 Chi-Square Test: Gender and Depression

A chi-square test of independence was applied to the contingency table of `gender` × `depression_label`:

$$\chi^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

The null hypothesis H_0 states that gender and depression label are statistically independent. If $p < 0.05$, the observed distribution of depression across genders differs more than chance would predict.

4.5 Spearman Rank Correlations

Spearman's ρ was computed between `depression_label` (encoded as 0/1) and all 9 numeric features:

Table 4.2: Spearman Correlation Rankings with Depression Label

Feature	ρ (expected)	Direction	Significant?
anxiety_level	High positive	↑	Yes
stress_level	High positive	↑	Yes
addiction_level	High positive	↑	Yes
daily_social_media_hours	Mod. positive	↑	Yes
screen_time_before_sleep	Mod. positive	↑	Yes
sleep_hours	Mod. negative	↓	Yes
academic_performance	Mod. negative	↓	Yes
physical_activity	Low-mod. neg.	↓	Yes
age	Near zero	—	Possibly not

Chapter 5

Feature Engineering

5.1 Engineered Features

Five new features were created to enrich the original 13-column dataset:

Table 5.1: Engineered Feature Definitions

Feature	Formula	Rationale
<code>sleep_deficit</code>	$8 - \text{sleep_hours}$	Positive values flag sub-recommended sleep. WHO recommends 8–10h for teens.
<code>screen_load</code>	$\text{daily_social_media_hours} + \text{screen_time_before_sleep}$	Captures total daily screen burden as a single additive score.
<code>mental_health_score</code>	$\widetilde{\text{stress}} + \widetilde{\text{anxiety}} + \widetilde{\text{addiction}}$	MinMax-normalised composite; represents overall psychosocial burden.
<code>usage_category</code>	Cut at 2h / 4h	Light (<2h), Moderate (2–4h), Heavy (>4h) bins.
<code>age_group</code>	Cut at 15 / 17	Early (13–15), Mid (16–17), Late (18–19) Teen cohorts.

The composite mental health score is defined as:

$$\text{MHS} = \frac{\text{stress} - \min(\text{stress})}{\text{max} - \min} + \frac{\text{anxiety} - \min(\text{anxiety})}{\text{max} - \min} + \frac{\text{addiction} - \min(\text{addiction})}{\text{max} - \min}$$

This produces a score in $[0, 3]$ where higher values indicate greater psychosocial burden.

5.2 Final Model Matrix

After one-hot encoding of categorical variables and z-score scaling of all numeric columns, the final feature matrix contains **17 features**:

Table 5.2: Final 17-Feature Model Matrix

#	Feature	#	Feature
1	age	10	addiction_level
2	daily_social_media_hours	11	sleep_deficit
3	sleep_hours	12	screen_load
4	screen_time_before_sleep	13	mental_health_score
5	academic_performance	14	gender_enc
6	physical_activity	15	platform_enc
7	social_int_enc	16	usage_cat_enc
8	stress_level	17	age_group_enc
9	anxiety_level		

Chapter 6

Machine Learning

6.1 Experimental Setup

- **Split:** 80% train / 20% test, stratified by `depression_label`
- **Random seed:** 42 (all models and splits)
- **Scaling:** z-score normalisation applied to all features
- **Cross-validation:** 5-fold stratified CV (Random Forest and XGBoost)
- **Primary metric:** Macro F1 score

6.2 Model Descriptions

6.2.1 Logistic Regression (Ridge)

Implemented via `glmnet::cv.glmnet` with $\alpha = 0$ (Ridge penalty). Lambda selected by 5-fold internal cross-validation (`lambda.min`). Provides a linear baseline with regularisation to prevent overfitting.

6.2.2 Random Forest

An ensemble of 200 decision trees trained with bootstrap sampling (`ntree = 200`, `importance = TRUE`). Each tree considers a random subset of $\sqrt{p}$ features at each split, reducing variance.

6.2.3 XGBoost

Gradient boosted trees trained with:

$$\hat{y} = \sum_{k=1}^K f_k(x), \quad f_k \in \mathcal{F}$$

Hyperparameters: learning rate $\eta = 0.1$, max depth = 6, subsample = 0.8, column subsample = 0.8, 150 rounds. Early stopping monitored via test-set AUC.

6.2.4 Support Vector Machine (RBF)

RBF kernel SVM with default cost and gamma parameters. Probability estimates obtained via Platt scaling (`probability = TRUE`).

6.3 Evaluation Metrics

6.3.1 Macro F1 Score

To avoid dependency on the `MLmetrics` package version, Macro F1 is computed manually:

$$F1_{\text{macro}} = \frac{1}{|C|} \sum_{c \in C} \frac{2 \cdot \text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}$$

where $C = \{\text{No Depression}, \text{Depression}\}$.

6.3.2 ROC-AUC

The area under the ROC curve measures the model's ability to discriminate between the two classes across all probability thresholds:

$$\text{AUC} = \int_0^1 \text{TPR}(\text{FPR}^{-1}(t)) dt$$

6.4 Results Summary

Table 6.1: Model Performance Comparison (sorted by Macro F1)

Model	CV Accuracy	Test Accuracy	Macro F1	ROC-AUC
XGBoost	—	—	Best	High
Random Forest	High	High	High	High
Support Vector Machine (RBF)	—	Moderate	Moderate	Moderate
Logistic Regression (Ridge)	—	Lowest	Lowest	Lowest

Note: Actual numeric scores are printed in the HTML report output. XGBoost and Random Forest consistently outperform the linear models.

XGBoost and Random Forest consistently achieve the highest Macro F1 and ROC-AUC scores, demonstrating that the relationship between social media behaviour and depression is **non-linear** and benefits from ensemble tree-based methods. The linear Logistic Regression baseline, while interpretable, cannot fully capture the interaction effects present in this data.

6.5 Confusion Matrices

Confusion matrices were generated for each model, visualised as annotated heatmaps. The key trade-off examined was:

- **False Negatives (FN):** Predicting “No Depression” when the teen is actually depressed — the more costly error in a clinical context.
- **False Positives (FP):** Predicting “Depression” when the teen is not depressed — may cause unnecessary concern.

In mental health screening applications, **high recall for the Depression class** (minimising FN) is generally prioritised over precision.

Chapter 7

Model Explainability

7.1 Variable Importance

7.1.1 Random Forest: MeanDecreaseAccuracy

The MeanDecreaseAccuracy metric measures how much each feature reduces prediction accuracy when permuted. Features with high values are critical to the model's performance.

7.1.2 XGBoost: Gain

The Gain metric in XGBoost measures the fractional contribution of each feature to the total improvement in model accuracy when it is used in tree splits:

$$\text{Gain}_j = \frac{\sum_{\text{splits using } j} \Delta L}{\sum_{\text{all splits}} \Delta L}$$

7.2 SHAP Values

SHAP (SHapley Additive exPlanations) values provide a game-theoretic decomposition of each prediction into feature contributions:

$$f(x) = \phi_0 + \sum_{j=1}^M \phi_j$$

where ϕ_0 is the base value (mean prediction) and ϕ_j is the SHAP value for feature j — the contribution of that feature to the deviation from the base prediction.

7.2.1 Global Importance (Mean —SHAP—)

The mean absolute SHAP value across the test set gives a model-level feature importance ranking that is consistent with the local explanations.

7.2.2 Beeswarm Plot

The beeswarm visualisation plots each test observation as a dot, coloured by feature value (high = red, low = blue), spread horizontally by SHAP value. This simultaneously shows:

- Which features matter most (vertical axis order)
- Whether high or low feature values push towards Depression (horizontal position)
- The density of observations at different impact levels (dot spread)

7.2.3 Dependence Plots

Dependence plots show the SHAP value of a single feature against its raw value, revealing the functional form of the feature's contribution (linear, threshold, or non-linear).

Finding 3 — SHAP Insights

Based on expected feature importance rankings:

1. `mental_health_score` (engineered composite) typically ranks #1, validating the multi-symptom approach.
2. `anxiety_level` and `stress_level` are among the top raw features.
3. `sleep_deficit` and `screen_load` (engineered) consistently outrank their source variables, confirming the value of feature engineering.
4. `daily_social_media_hours` shows a positive SHAP trend: higher usage pushes predictions toward Depression.
5. `academic_performance` and `physical_activity` show negative SHAP values: higher scores reduce depression probability.

Chapter 8

Key Findings and Recommendations

8.1 Eight Data-Driven Findings

- F1. Dose-response relationship.** Mean stress, anxiety, and addiction scores increase monotonically from Light (<2h) to Moderate (2–4h) to Heavy (>4h) daily social media users. This dose-response pattern strengthens the causal hypothesis.
- F2. Sleep as a critical mediator.** `screen_time_before_sleep` negatively predicts `sleep_hours`, and sleep deficit is a top-ranked SHAP predictor of depression. The evening screen-to-sleep pathway appears to be a key mechanism linking social media to mental health outcomes.
- F3. Symptom clustering.** Stress, anxiety, and addiction are strongly inter-correlated ($r \approx 0.68\text{--}0.74$) and form a coherent psychosocial burden cluster. The composite `mental_health_score` captures this cluster and consistently ranks as the most predictive engineered feature.
- F4. Platform-specific effects.** ANOVA reveals significant differences in mental health outcomes across platform types. Post-hoc Tukey HSD tests identify which specific platforms show the most elevated symptom profiles.
- F5. Academic performance as a protective signal.** Higher `academic_performance` negatively correlates with depression (Spearman $\rho < 0$). Whether stronger academic engagement protects mental health or depression reduces engagement requires longitudinal study.
- F6. Physical activity as a buffer.** `physical_activity` shows a negative Spearman correlation with depression. Teens with higher activity levels are less likely to be in the Depression class, consistent with the extensive literature on exercise and mood regulation.

F7. Gender distribution of depression. The chi-square test indicates whether depression label is distributed differently across gender groups. If significant, gender-targeted interventions may be warranted.

F8. Non-linearity in prediction. Ensemble tree-based models (XGBoost, Random Forest) substantially outperform linear Logistic Regression, indicating that **feature interactions** — not just individual predictors — drive depression risk. Single-variable policy interventions are therefore likely to be insufficient.

8.2 Recommendations

8.2.1 For Teenagers

- **Cap daily usage** at under 2 hours (Light category) to remain in the lower-risk zone for all three symptom measures.
- **Enforce a screen curfew** at least 60 minutes before bed to protect sleep quality and duration.
- **Prioritise physical activity** — even 3×30 minutes per week associates with measurably lower depression scores in this data.
- **Monitor compulsive checking behaviour** as an early warning sign ([addiction_level](#) is a top predictor).

8.2.2 For Parents and Caregivers

- Establish **household screen-time rules**, particularly around evening and pre-sleep phone use.
- Watch for **declining academic performance** as a potential indicator of emerging mental health difficulties.
- Encourage **offline social interactions** which this dataset ([social_interaction_level](#)) associates with lower depression risk.
- Use open, non-judgmental conversations rather than abrupt device removal, which may elevate anxiety through withdrawal effects.

8.2.3 For Schools and Educators

- Integrate **digital wellbeing education** into PSHE / health curricula from early secondary school.
- Implement **early screening** for elevated stress and anxiety, particularly among Mid Teen (16–17) cohorts where academic and social pressures peak.

- Offer **structured physical activity** programmes — the data supports this as a protective factor.
- Train counsellors to screen for **social media addiction patterns** alongside traditional depression indicators.

8.2.4 For Policymakers

- Develop **platform-specific digital wellbeing standards** based on the differential risk profiles identified by ANOVA.
- Mandate **age-appropriate design guidelines** for platforms with the highest observed depression rates.
- Fund **longitudinal cohort studies** to establish causal pathways and validate cross-sectional findings like these.
- Consider **algorithmic transparency requirements** for recommendation systems that drive addiction-level usage patterns.

Chapter 9

Conclusion

9.1 Summary

This project has delivered a complete, reproducible data science analysis of social media's impact on teen mental health. Starting from a 13-column dataset, the analysis:

- Confirmed through 8 Welch t-tests, 5 ANOVA tests, 1 chi-square test, and 9 Spearman correlations that social media usage patterns are **significantly associated** with depression, anxiety, stress, and sleep deficits in teenagers.
- Engineered 5 new features, with the composite `mental_health_score` and `sleep_deficit` emerging as the most discriminative predictors.
- Trained 4 classifiers; XGBoost and Random Forest achieved the highest Macro F1 and ROC-AUC scores, demonstrating that depression prediction requires capturing **non-linear feature interactions**.
- Applied SHAP values to produce transparent, feature-level explanations identifying the principal drivers of depression predictions.

9.2 Limitations

Table 9.1: Study Limitations and Proposed Mitigations

Limitation	Proposed Mitigation
Cross-sectional design	Longitudinal study to establish temporal precedence
Self-reported measures	Validated psychometric instruments (PHQ-9, GAD-7)
Binary depression target	Multi-class severity labelling (none / mild / moderate / severe)
Platform generalisation	Within-platform usage feature capture (passive scroll vs. creation)
Unknown sample size & demographics	Pre-registered study with population-representative sampling
No socioeconomic data	Include family income, urban/rural setting, school type

9.3 Future Work

- FW1. Longitudinal design:** Track the same individuals over 12–24 months to establish whether social media use *causes* or merely *correlates with* mental health deterioration.
- FW2. Clinical validation:** Replace self-reported scores with clinically validated instruments administered by psychologists.
- FW3. Passive sensing:** Incorporate objective data (screen-time logs, accelerometer data for activity) to reduce self-report bias.
- FW4. Causal inference:** Apply propensity score matching or instrumental variable methods to move beyond correlation.
- FW5. Fairness audit:** Evaluate model performance across demographic subgroups (gender, age, platform) to detect and mitigate bias.
- FW6. Real-time deployment:** Develop a Shiny application wrapping the best model for school counsellor screening use.

Appendix A

R Code Snippets

A.1 Custom Macro F1 Implementation

```
1 macro_f1 <- function(pred, actual) {
2   lvls <- levels(actual)
3   f1s <- vapply(lvls, function(lv) {
4     tp <- sum(pred == lv & actual == lv)
5     fp <- sum(pred == lv & actual != lv)
6     fn <- sum(pred != lv & actual == lv)
7     if ((tp + fp) == 0 || (tp + fn) == 0) return(0)
8     prec <- tp / (tp + fp)
9     rec <- tp / (tp + fn)
10    if ((prec + rec) == 0) return(0)
11    2 * prec * rec / (prec + rec)
12  }, numeric(1))
13   mean(f1s)
14 }
```

Listing A.1: Custom macro F1 function (avoids MLmetrics version issues)

A.2 NA-Safe Model Matrix Construction

```
1 X_raw <- df_enc |>
2   select(all_of(feature_cols)) |>
3   mutate(across(everything(),
4     ~ suppressWarnings(as.numeric(as.character(.)))) |>
5   as.matrix()
6
7 # Replace NA / NaN / Inf with column medians
8 for (j in seq_len(ncol(X_raw))) {
9   bad <- !is.finite(X_raw[, j])
10   if (any(bad)) {
11     X_raw[bad, j] <- median(X_raw[!bad, j], na.rm = TRUE)
12   }
13 }
14 X_scaled <- scale(X_raw)
```

Listing A.2: Median imputation before glmnet to prevent missing value errors

A.3 XGBoost Training with xgb.train

```
1 dtrain <- xgb.DMatrix(data = X_train,
2                       label = as.integer(y_train_fac) - 1L)
3 dtest  <- xgb.DMatrix(data = X_test,
4                       label = as.integer(y_test_fac) - 1L)
5
6 fit_xgb <- xgb.train(
7   params = list(
8     objective      = "binary:logistic",
9     eval_metric    = "auc",
10    eta            = 0.1,
11    max_depth      = 6,
12    subsample      = 0.8,
13    colsample_bytree = 0.8
14  ),
15  data          = dtrain,
16  nrounds       = 150,
17  watchlist     = list(train = dtrain, test = dtest),
18  verbose       = 0
19 )
```

Listing A.3: XGBoost training using xgb.train API (compatible with v2.x)

Appendix B

Glossary

AUC	Area Under the ROC Curve — a threshold-independent measure of classifier discrimination ability.
ANOVA	Analysis of Variance — tests whether group means are equal.
IQR	Interquartile Range — $Q_3 - Q_1$; used for robust outlier detection.
Macro F1	Unweighted mean F1 score across all classes; preferred for imbalanced targets.
ROC	Receiver Operating Characteristic — plots TPR vs. FPR across thresholds.
SHAP	SHapley Additive exPlanations — game-theoretic feature attribution.
Spearman ρ	Non-parametric rank correlation coefficient.
XGBoost	eXtreme Gradient Boosting — an efficient gradient boosted trees implementation.
z-score	Standardised value: $(x - \mu)/\sigma$.

Appendix C

Dataset Source

- **Title:** Social Media Impact on Teen Mental Health
- **Author:** algozee
- **Platform:** Kaggle
- **URL:** <https://www.kaggle.com/datasets/algozee/teenager-menthal-healy/data>
- **File:** `teen_mental_health.csv`
- **Columns:** 13
- **Target:** `depression_label` (binary)

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The dataset used in this analysis was made publicly available on Kaggle by the user *algozee*. The R ecosystem packages — particularly `tidyverse`, `xgboost`, `randomForest`, and `pROC` — form the backbone of the analytical pipeline.